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AI Applications and Case Histories — ITAI 2372

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20 June 2026

A04: AI-Driven Fraud Detection – Cognizant Case Study

Introduction

Fraud detection is one of the most significant challenges facing financial institutions today. Banks process millions of transactions daily and lose billions of dollars each year to fraudulent activities, including counterfeit checks. Traditional fraud detection methods often depend on manual verification processes, requiring employees to review transactions individually. While these methods can identify suspicious activity, they are time-consuming, costly, and difficult to scale.

Cognizant is a multinational technology and professional services company specializing in digital transformation, artificial intelligence, cloud computing, and business consulting solutions. The company serves clients across multiple industries, including healthcare, manufacturing, retail, and financial services. Cognizant maintains dedicated Artificial Intelligence and Banking & Financial Services practices, allowing it to develop specialized solutions for complex financial challenges such as fraud detection.

To address these challenges, Cognizant developed an AI-driven fraud detection solution for a global banking client. By leveraging machine learning, neural networks, deep learning, and optical character recognition (OCR), the system automates check verification and identifies potentially fraudulent transactions in real time. This case study examines the technology behind the solution, the benefits achieved, and the challenges associated with implementing AI-powered fraud detection in the banking industry.

Technology Overview

Cognizant's fraud detection solution combines multiple artificial intelligence technologies to improve the speed and accuracy of check verification. The system was built using Google TensorFlow, an open-source machine learning framework that supports the development and deployment of AI models. The process begins when digital images of checks are analyzed using OCR technology. OCR extracts important information such as signatures, payee names, routing numbers, check numbers, and other transaction details. This information is then converted into structured data that can be evaluated by the AI model.

The extracted data is then processed by a neural network trained on historical check portfolios and known fraud patterns. Using machine learning and deep learning techniques, the system identifies anomalies and suspicious characteristics that may indicate fraudulent activity. Instead of making a simple approve-or-reject decision, the model generates a confidence score that estimates the likelihood of fraud. Transactions with higher risk scores can then be reviewed by fraud specialists for further investigation.

The system demonstrates significant operational performance. According to Cognizant, the model can process up to 1,200 checks per second while generating fraud confidence scores in less than 70 milliseconds. This allows financial institutions to identify suspicious transactions almost instantly while maintaining high processing volumes.

Benefits

The implementation of AI significantly improved the bank's fraud detection capabilities. During testing on historical portfolios, Cognizant reported a 50 percent reduction in fraudulent transactions. This result demonstrates the effectiveness of machine learning in identifying patterns that may be difficult to detect through traditional manual review processes.

The solution also generated substantial financial benefits. Cognizant projected approximately \$20 million in annual savings through reduced fraud losses. By identifying suspicious checks before losses occur, the bank can better protect its assets while reducing financial risk.

Operational efficiency also improved considerably. Manual fraud detection often requires large teams of employees to review transactions one by one, creating delays and increasing costs. The AI solution automates much of this process, allowing fraud analysts to focus their attention on high-risk cases instead of routine transaction reviews.

In addition, the system offers improved scalability and consistency. Unlike human reviewers, AI models can analyze large volumes of transactions continuously while applying the same evaluation criteria to every check. This helps maintain a more reliable fraud detection process as transaction volumes increase.

Challenges

Despite its advantages, implementing AI for fraud detection presents several challenges. One of the most important technical challenges is the need for high-quality training data. Machine learning models rely on large amounts of historical data to learn fraud patterns effectively. If the data is incomplete, inaccurate, or outdated, model performance may suffer.

Another challenge involves balancing false positives and false negatives. A false positive occurs when a legitimate transaction is incorrectly flagged as fraudulent, potentially causing inconvenience for customers. A false negative occurs when fraudulent activity goes undetected. Financial institutions must continuously monitor and refine their models to maintain an appropriate balance between security and customer experience.

Operational challenges must also be considered. Integrating AI solutions into existing banking systems can require significant investments in technology infrastructure, employee training, and process redesign. Employees must understand how to interpret AI-generated recommendations and determine when additional investigation is necessary.

Finally, AI-based fraud detection raises important concerns regarding privacy and transparency. These systems rely on large volumes of customer and transaction data, making data protection a critical responsibility. In addition, complex machine learning models can make it difficult to fully explain why a particular transaction was flagged as suspicious. Maintaining human oversight and clear decision-making processes remains essential for building trust in AI-driven financial systems.

Conclusion

Cognizant's AI-driven fraud detection solution demonstrates how artificial intelligence can improve fraud prevention within the banking industry. By combining OCR, machine learning, neural networks, and deep learning, the system can analyze checks in real time, identify suspicious patterns, and significantly reduce fraudulent transactions.

The results reported by Cognizant — including a 50 percent reduction in fraudulent transactions, projected annual savings of \$20 million, response times below 70 milliseconds, and the ability to process up to 1,200 checks per second — highlight the practical value of AI in financial services.

This case study illustrates that AI can enhance both the effectiveness and efficiency of fraud detection when supported by high-quality data, ongoing model monitoring, and appropriate human oversight. For financial institutions facing increasingly sophisticated fraud threats, AI represents a powerful tool for strengthening security while improving operational performance.

References:

Cognizant. (n.d.). *AI-driven solution reduces fraud risk for bank*.

Cognizant. (n.d.). *AI machine learning aids fraud detection*. Retrieved from Cognizant case study website.

Author's Note:

Google NotebookLM and ChatGPT were used as learning and research support tools throughout the development of this assignment. These tools assisted with organizing course materials, exploring concepts, and deepening understanding of topics discussed in class, including AI advancements, ethics, governance, and future trends. They were used to support the learning process rather than as primary sources of information.